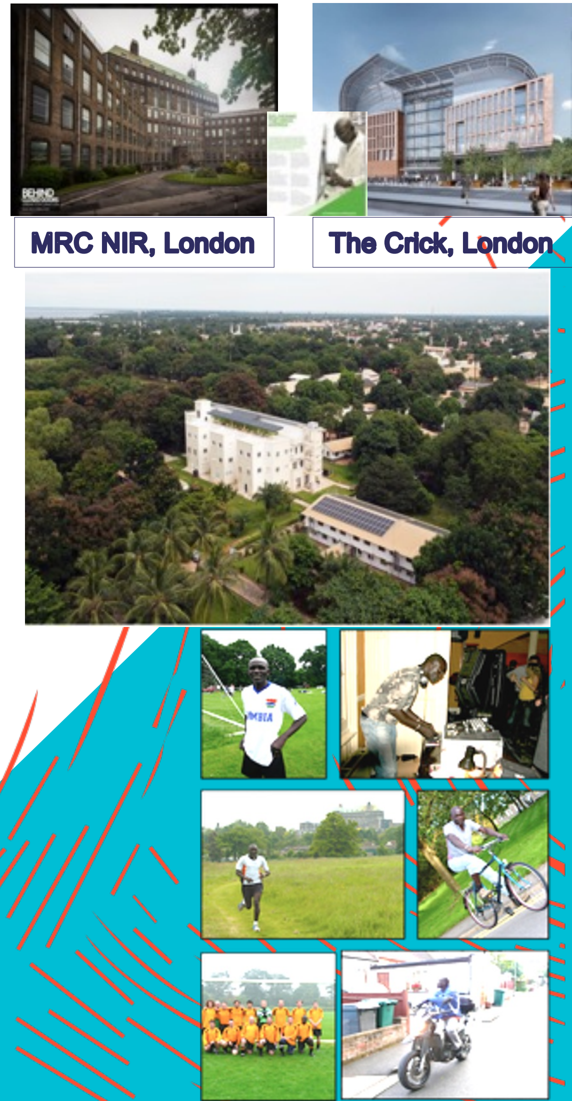
Introduction to Genomics Technologies



Wellcome Connecting Science
13th September 2025

Abdul Karim Sesay
Head of Genomics Strategic Core Platform

UKRI
MRC Unit The Gambia
LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

Outline of presentation

- The Human Genome Project
- Outcome from the HGP
- Sanger Sequencing versus High-throughput Sequencing
- Next Generation Sequencing methodologies
- Establishing the Genomics Core facility at the MRCG at LSHTM

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History and future of DNA sequencing

- 1953: Discovery of DNA structure by Watson and Crick
- 1967: First DNA sequence of 11 bp published (20 pages)
- 1976: First genome sequenced: Bacteriophage MS2 (3569 bp) by Walter Fiers
- 1977: Sanger sequencing method published
- 1980: Nobel Prize Wally Gilbert and Fred Sanger
- 1982: Genbank started
- 1983: development of PCR
- 1987: 1st automated sequencer: Applied Biosystems Prism 373
- 1996: Capillary sequencer: ABI 310
- 1998: Genome of Caenorhabditis elegans sequenced (100 million bp)
- 2001: Human genome sequenced (3,2 billion bp)
- 2005: 1st 454 Life Sciences Next Generation Sequencing system: GS 20 system († mid 2016)
- 2006: 1st Solexa Next Generation Sequencer: Genome Analyzer (Illumina)
- 2007: 1st Applied Biosystems Next Generation Sequencer: SOLiD († Dec 2017)
- 2009: 1st Helicos single molecule sequencer: Helicos Genetic Analyser System († Nov 2012)
- 2011: 1st Ion Torrent Next Generation Sequencer: PGM
  1st Pacific Biosciences single molecule sequencer: PacBio RS Systems
- 2012: Oxford Nanopore Technologies demonstrates ultra long single molecule reads
- 2014: Roche acquires Genia: development of NanoTag single molecule sequencing
- 2015: 1st BGI Next Generation Sequencer: BGISEQ-500 (sold in China only)
- 2016: 1st Oxford Nanopore Technologies sequencer: MinION
- 2017: SeqLL announces tSMS sequencer: single molecule (Helicos technology)


If it cost $3 billion to sequence
the human genome originally
– how on earth
can we imagine doing it
for less than $2,000 now?
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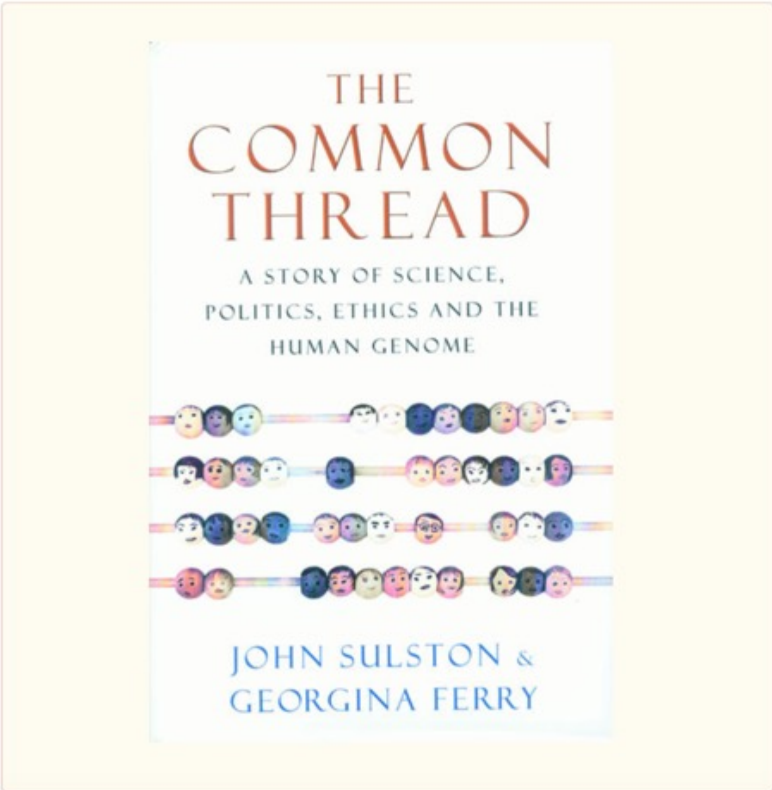
The automated production line for Sanger sequencing

The Race to Sequence the Human Genome



The book retraces the decision by a few individuals to start such a bold project, the ensuing battles to obtain funding from governments throughout the world and the panic caused by the sudden announcement that a private company was entering the race

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Journal and magazine covers celebrating the sequencing of the human genome

In 2003 the largest ever collaborative project in biology was declared complete. Scientists from the US, the UK, Germany, France, Japan and China had deciphered a majority of the human genome — the DNA blueprint present in every human cell.

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What have we learned so far from HGP?

https://www.sanger.ac.uk/HGP/

Diseases that affect large numbers of people are complex and have many different genetic contributing factors (or susceptabilities).

Even when we know the exact genetic cause of a disease there may well be nothing we can currently do about it.

Finding the causes of complex diseases

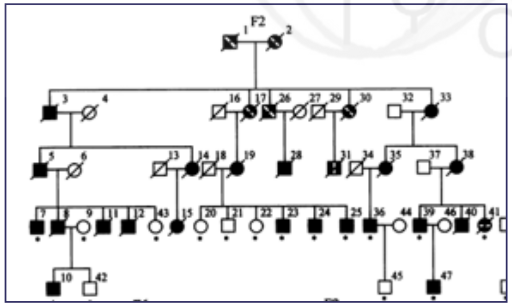
With any disease that has a hereditary component, we want to find the genetic elements responsible.

Monogenic disorders

These are due to mutation(s) in a single gene, and have been easy to find, using families where the disease appears (linkage studies). For example cystic fibrosis, where commonest mutation, ΔF508, causes the loss of a single amino acid (phenylalanine at position 508).

Complex diseases

Such as type-II diabetes, have a clear hereditary component, but are also strongly influenced by environmental cues like body weight. Attempts to find them by studying affected family trees were spectacularly unsuccessful.



M Nyström-Lahti et al. Proc Natl Acad Sci U S A. 1994 June 21; 91(13): 60546058

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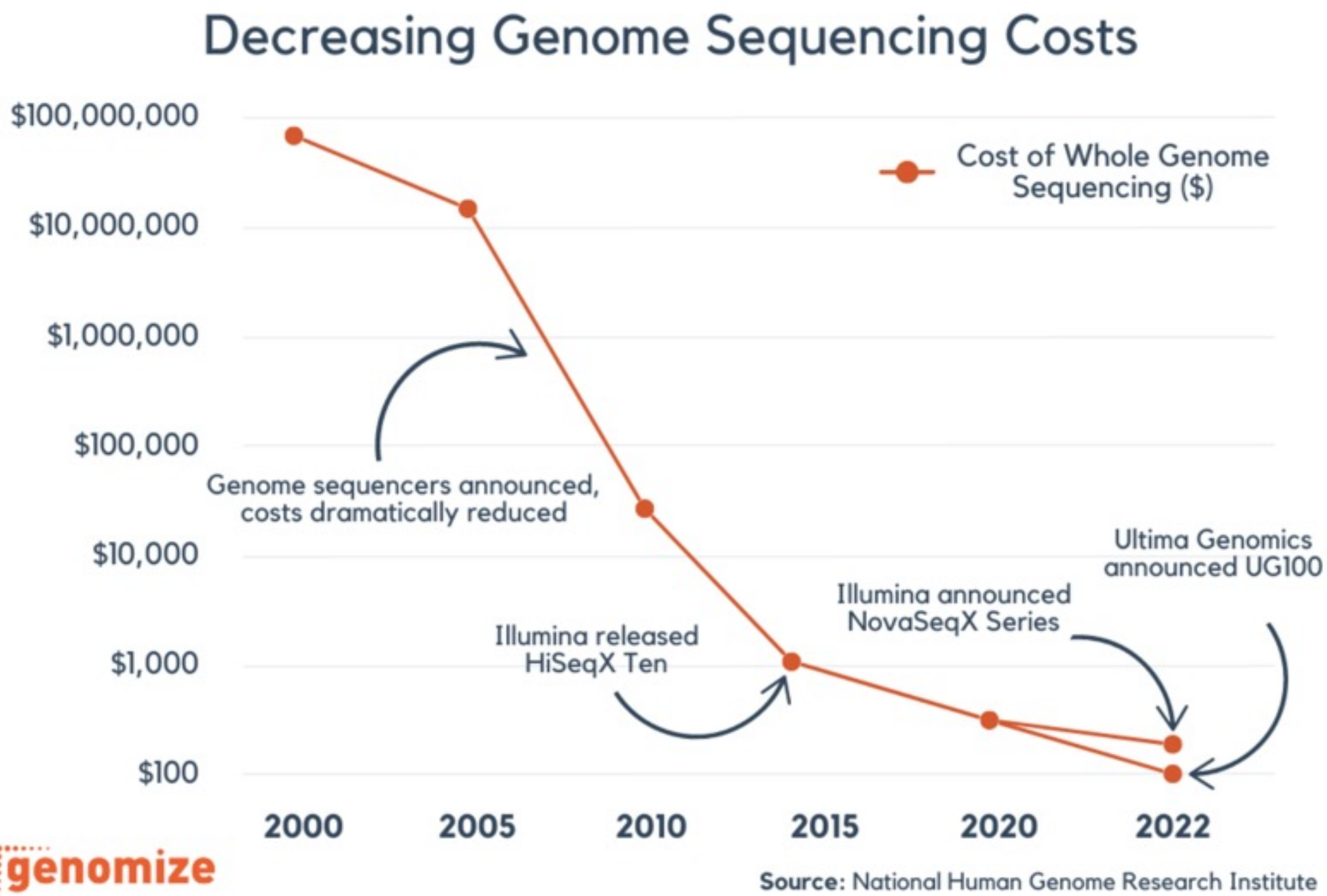
Outcomes from Human Genome Project

The Human Genome Project has already begun to have a major impact on the life sciences and the quality of our lives.

- Molecular medicine
- New uses for microbes.
- Risk assessment
- Comparative genomics
- Forensic science
- Better crops and animals
- Personalised healthcare

Although it has proven to be a highly successful effort and will certainly achieve all of its stated goals, it really marks only the most basic of beginnings in understanding the genetic secrets of life.



genomize

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Different names for the same thing…

- Second generation sequencing
- Next generation sequencing (NGS)
- Ultra high-throughput sequencing
- Sequencing-by synthesis
- Parallel sequencing
- Pyro-sequencing
- Short read sequencing
- Deep sequencing

Generally everything other than old style Sanger sequencing

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NGS and Sanger sequencing

- No library construction by cloning, novel PCR techniques used and fragments are separated by physico-chemical properties
- No need for the separation of elongation products, no capillary or gel lane, several (lots of) fragments are sequenced in parallel

Sanger DNA Sequencing Method (manual)

- Uses dideoxy nucleotides to terminate DNA synthesis
- DNA synthesis reactions in four different tubes
- Radioactive d ATP is also included in all the tubes so the DNA products will be radioactive
- Yielding a series of DNA fragments whose sizes can be measured by electrophorisis
- The last base of each fragments is known

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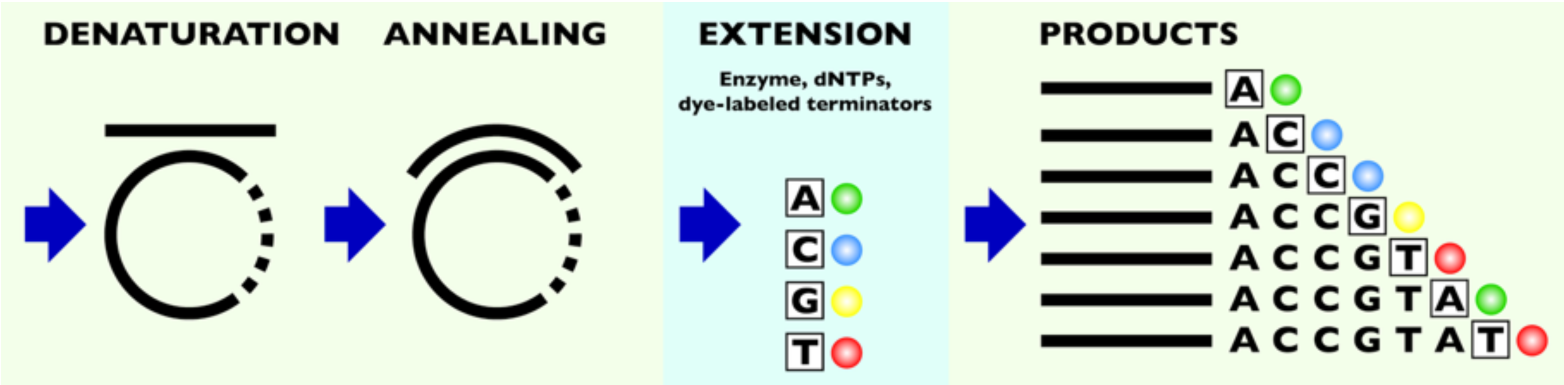
Automated DNA Sequencing



Cycle sequencing is a simple method in which successive rounds of denaturing, annealing and extension in a thermal cycler resulting in linear amplification of extension products. With dye terminators labeling , each of the four dideoxy terminators (ddNTPs) is tagged with a different fluorescent dye

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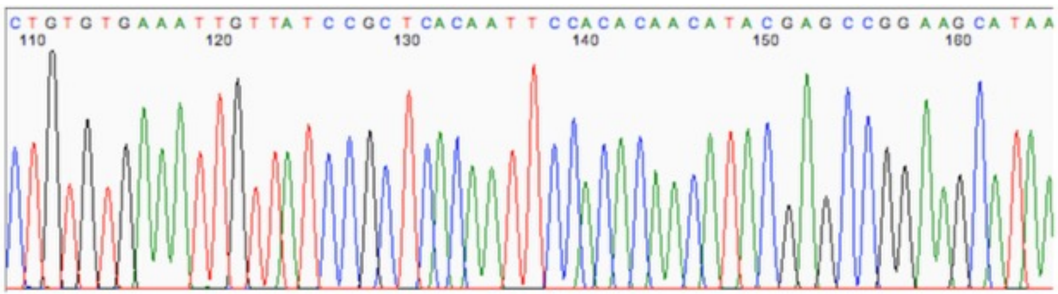
Automated DNA Sequencing

- The primer extension reactions are run the same way as the manual method
- Reaction is carried out in one tube and possible products are actually produced
- The various reaction products separate according to size on gel electrophoresis



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LAMP and qPCR

LAMP and qPCR are both nucleic acid amplification methods, but LAMP is an **isothermal (single temperature) technique** for rapid, qualitative detection, while qPCR is a **thermocycling technique** for quantitative analysis. LAMP is simpler, more accessible for low-resource settings, and faster for Yes/No results. qPCR is more versatile and suitable for applications requiring precise quantification and extensive molecular biology work.

NGS and qPCR

When comparing next-generation sequencing (NGS) vs quantitative PCR (qPCR) technologies, the key difference is discovery power. While both offer highly sensitive and reliable variant detection, qPCR can only detect known sequences. In contrast, NGS is a hypothesis-free approach that does not require prior knowledge of sequence information.

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LAMP, RCA, RPA, CRISPR, and NEAR are all nucleic acid amplification techniques used in molecular diagnostics and research.

They are all based on isothermal amplification, meaning they don't require temperature cycling like PCR.

These techniques can be used individually or in combination to improve the sensitivity and specificity of nucleic acid detection. The integration of these techniques has led to the development of rapid and sensitive POC diagnostic tools for a variety of applications.

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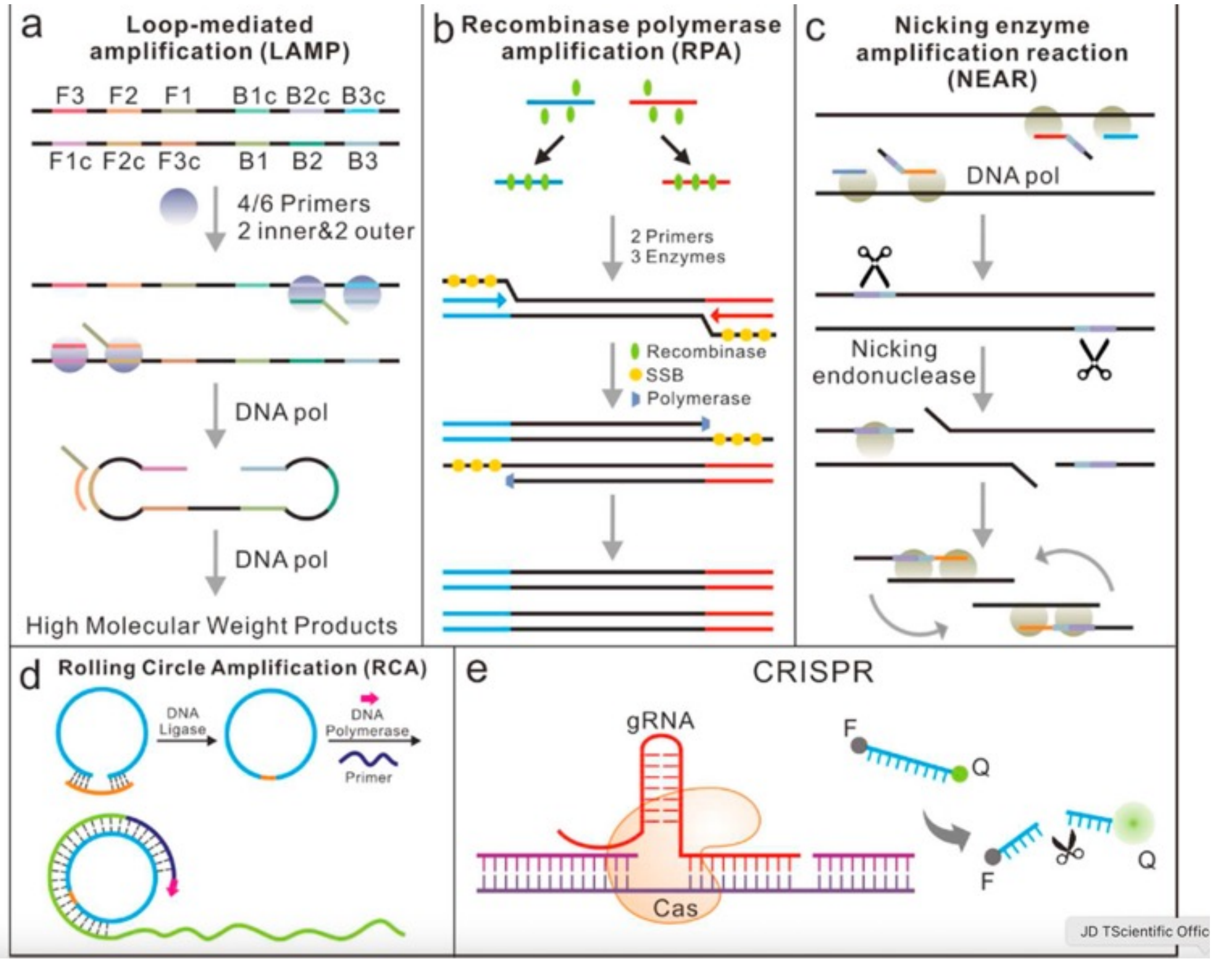

Microarray

Illumina Bead Array and Affymetrix Gene Chip

Microarray is a hybridization of a nucleic acid sample (target) to a very large set of oligonucleotide probes, which are attached to a solid support, to determine sequence or to detect variations in a gene sequence or expression or for gene mapping

Fluorescent in situ hybridisation (FISH)

Fluorescent in situ hybridisation is a cytogenetic technique used to identify and characterise chromosomal variants.

ACATTAGTTATACCGCGGCGGATTAATACAGTCGACCTTCGATCACCACAGACTTAG

TGTAATCAATATGGCGCCGCCTAATTATGTCAGCTGGAAGCTAGTGGTGTCTGAATC

Next Generation Sequencing methodologies

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NGS: Broadest Range of Applications

Human Genome Resequencing

Bacterial Sequencing | Targeted Resequencing

De novo Sequencing | ChIP-Seq | Small RNA Discovery

Tag profiling | mRNA-Seq | DNase I Hypersensitivity | Methylation

Molecular Cytogenetics | AFLP | ChIA-PET | CNV | Nucleosome Mapping

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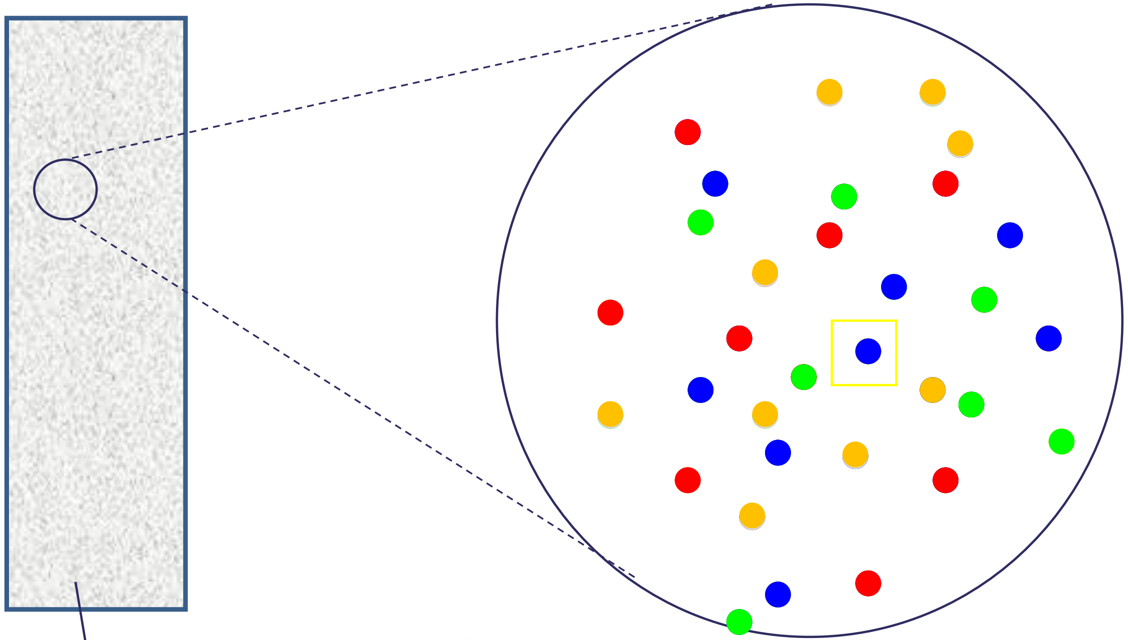
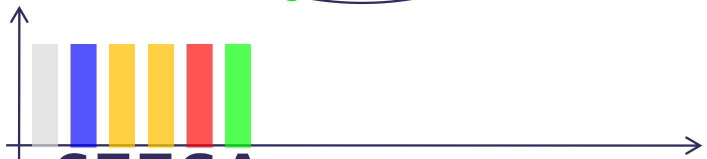
Illumina (Solexa) Next Generation Sequencing

The Genome Analyzer (GA), was developed by Solexa Inc, purchased by Illumina in 2007 for $650 million

Solexa founder, Cambridge University chemist
Shankar Balasubramanian



200 – 300 million colonies of different DNA strand



C T T G A . . . -> 150 bases

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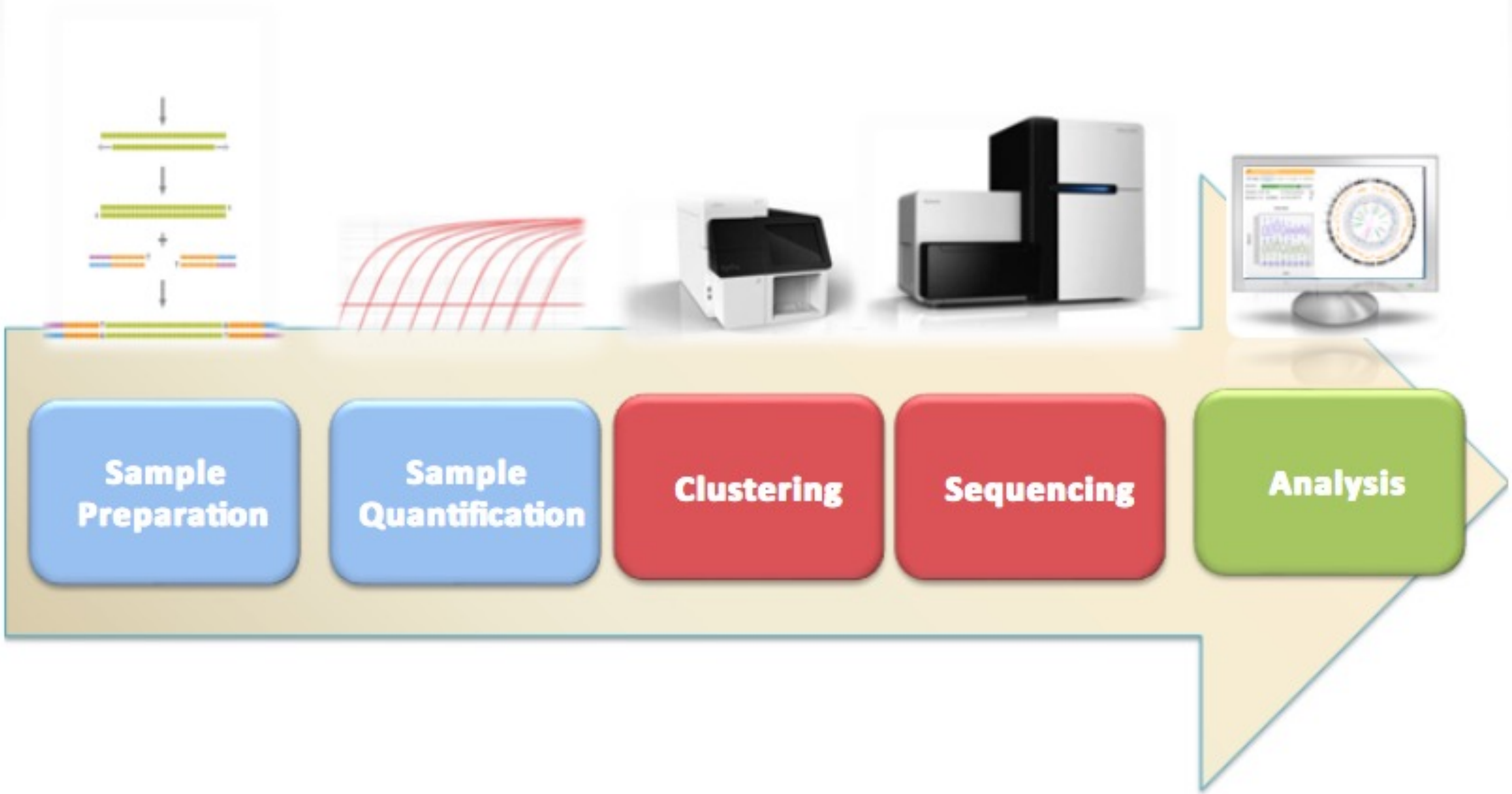
Illumina Workflow



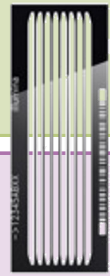
Illumina Sequencing Workflow – Sequencing

1 **Library Preparation**

- Fragment DNA
- Repair ends
- Add A overhang
- Ligate adapters
- Purify

2 **Cluster Generation**

- Hybridize to flow cell
- Extend hybridized template
- Perform bridge amplification
- Prepare flow cell for sequencing



3 **Sequencing**

- Perform sequencing
- Generate base calls

4 **Data Analysis**

- Images
- Intensities
- Reads
- Alignments

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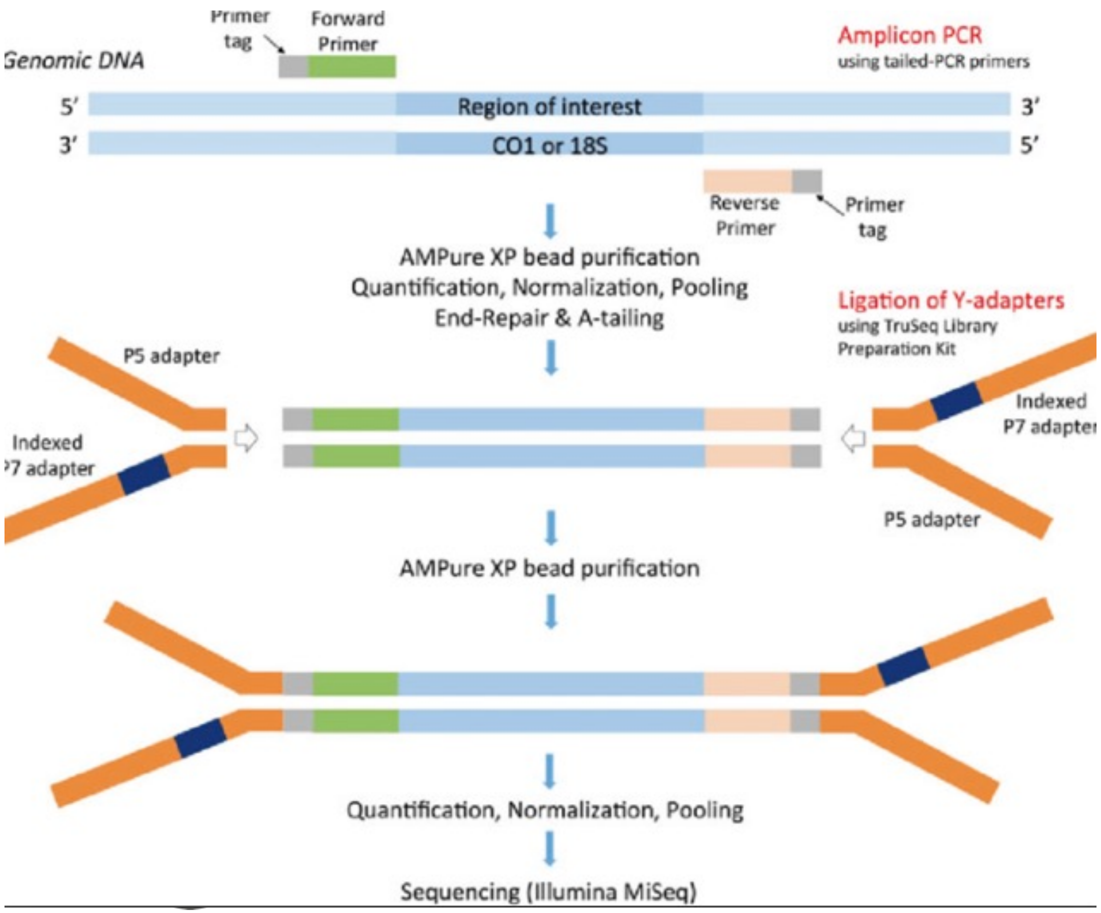
Example Sample Prep Workflow: TruSeq Paired-end Library

DNA RNA

Amplicon Library preparation



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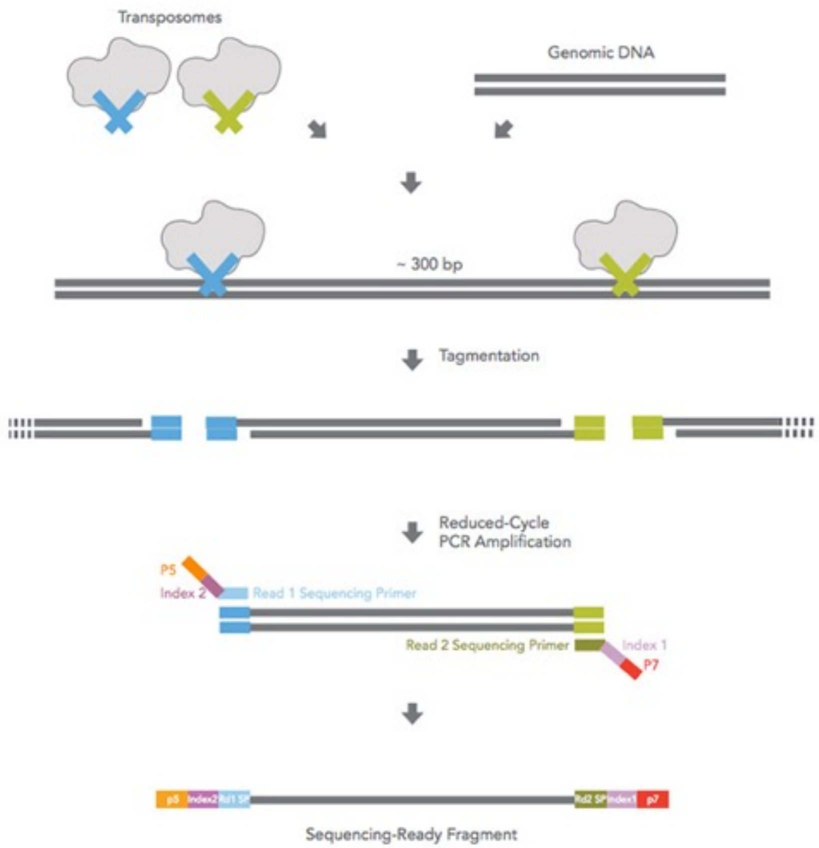
Nextera (transposons) Library preparation



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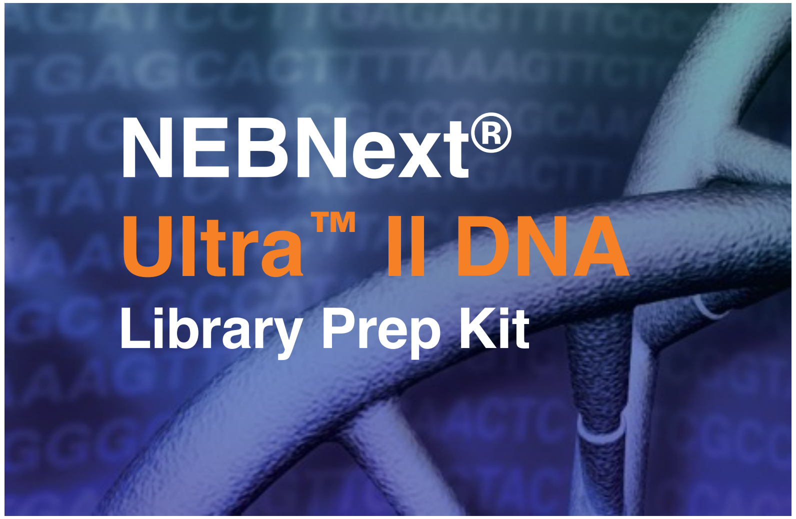


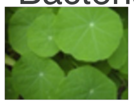
Broad Range of DNA Input

Genomic DNA

Tissue

Blood

Bacteria



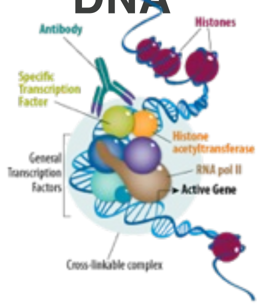
Plant

Virus

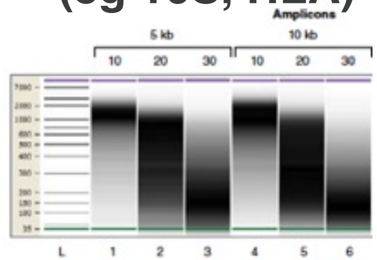
FFPE DNA
(Formalin-Fixed Parafin-Embedded)

Cell-free DNA
Extracted from liquid biopsies (e.g. blood, urine)

ChIP DNA



PCR Amplicons (eg 16S, HLA)



Input DNA: 500 pg – 1µg

Library Quality Control

Quality control of libraries is important

- Titration of flowcells and failed runs are expensive
- Try to identify issues before starting a sequencing run
- QC is specific to your samples

Quantification of libraries is important for accurate pooling and loading

- Concentration – Qubit
- Size profile – Agilent Tapestation
- Template quantification – QPCR (Kappa)

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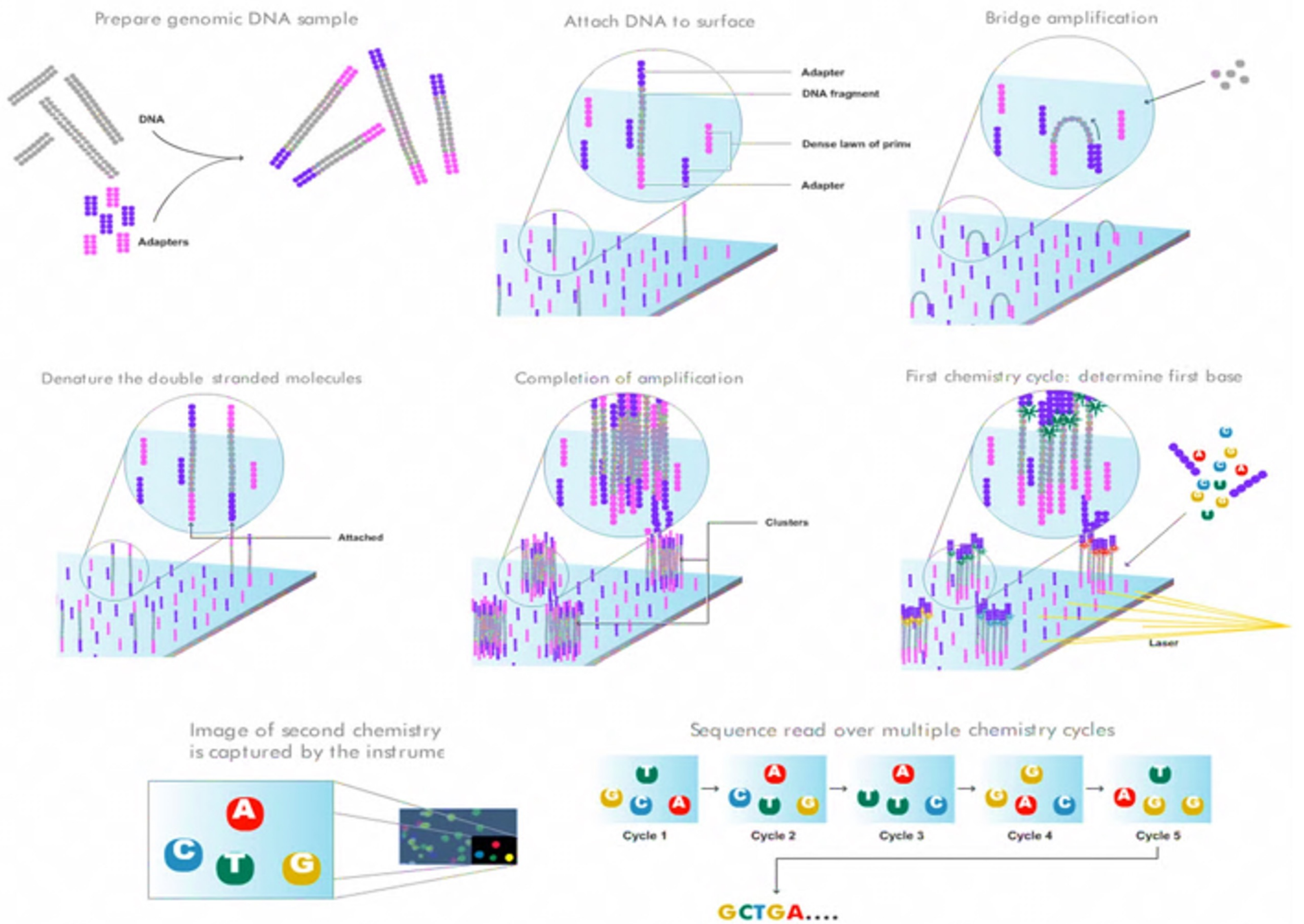
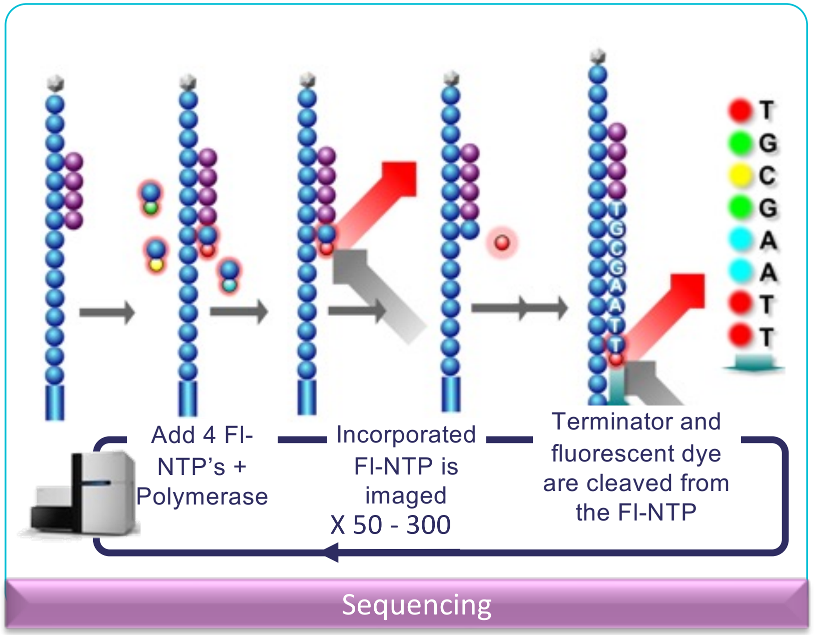

Prepare genomic DNA sample
DNA
Adapters
Attach DNA to surface
Adapter
DNA fragment
Dense lawn of prim
Adapter
Bridge amplification
Denature the double stranded molecules
Attached
Completion of amplification
Clusters
First chemistry cycle: determine first base
Laser
Image of second chemistry
is captured by the instrume
Sequence read over multiple chemistry cycles
Cycle 1
Cycle 2
Cycle 3
Cycle 4
Cycle 5
GCTGA....

Sequencing by synthesis (SBS technology)

- Same widely adopted and proven chemistry:
  - All four labeled nucleotides in one reaction
  - Higher accuracy
  - No problems with homopolymer repeats
- 3-step cycles:
  - Incorporate fluorescent nucleotide
  - Image tiles
  - Cleave terminator and fluorophores



MRCG NGS Sequencing Core:

Workflows, Platforms and Services

Process

Consultation and Design
Project Management
Project Submission
Costing and Billing
Reagent and Consumables
Sample Submission
Sample Quality Control
Library Preparation
Library Quality Control
Sequencing
Primary Data Analysis
Data Quality
Data Management and Storage
Sample Archiving

Equipment

Labelling machine Manual
Qubit Qiaxel Tapestation Glomax
Molecular equipment Manual and Automated
Qubit Qiaxel Tapestation Glomax Real-Time PCR
SeqStudio iSeQ MiSEQ Ion Torrent MinION NextSEQ 550 GridION PromethION NovaSEQ
Servers HPC
BioBank Freezers

Services

Quality Control, Pathogens WGS, Pathogens Transcriptomics, MLSD, Metagenomics, Targeted Methylation, Targeted Human Genome, Vector Transcriptomics, Host Transcriptomics, Host WGS, Vector WGS, GWAS, Whole Genome Methylation

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Long read technology comparisons

PacBio

Optical/fluorescence, High accuracy, kinetic analysis, long run times, very expensive

Oxford Nanopore Technologies

electrical current, Fast run time long read, inexpensive

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Oxford Nanopore Technologies - MinION

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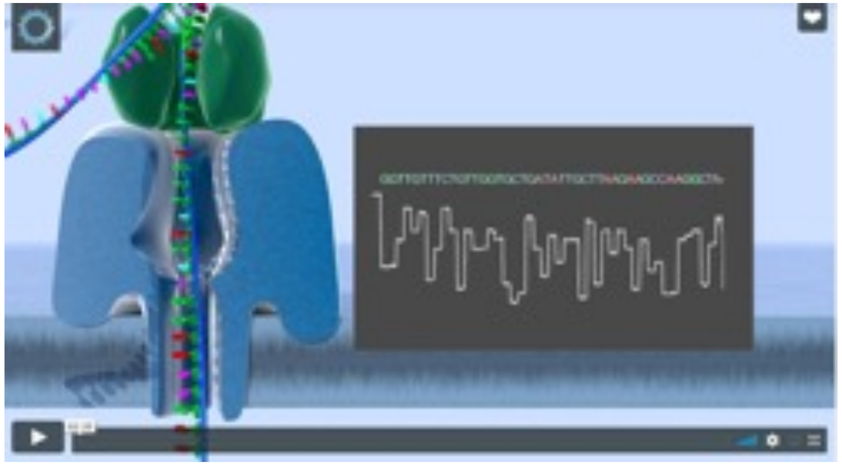
Why sequencing on a MinION

- Direct DNA/RNA sequencing
- REAL Real-time sequencing
- Ultra-long reads
- Scalable to portable or desktop
- 10 min library prep
- High yields for large genomes
- Easy Downstream (Cloud) analysis workflows



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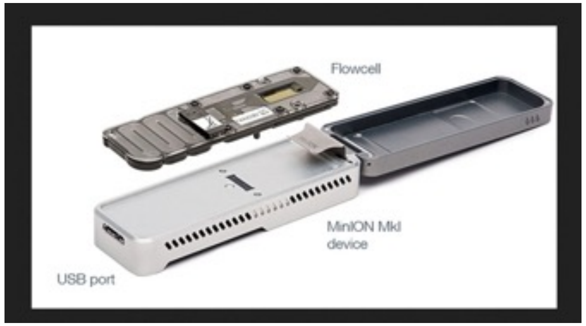
MinION Device and Flow Cell

- MinION Mk 1D device, released in April 2024
- SpotON Flow Cell Mk I (R10.4), 72 hr life /sequencing span.
- MinION the only portable, real time device for DNA and RNA, (protein) sequencing.
- Each consumable flow cell can now generate 15 Gb of DNA sequence data.
- The MinION streams data in real time so that analysis can be performed during the experiment and workflows are fully versatile.



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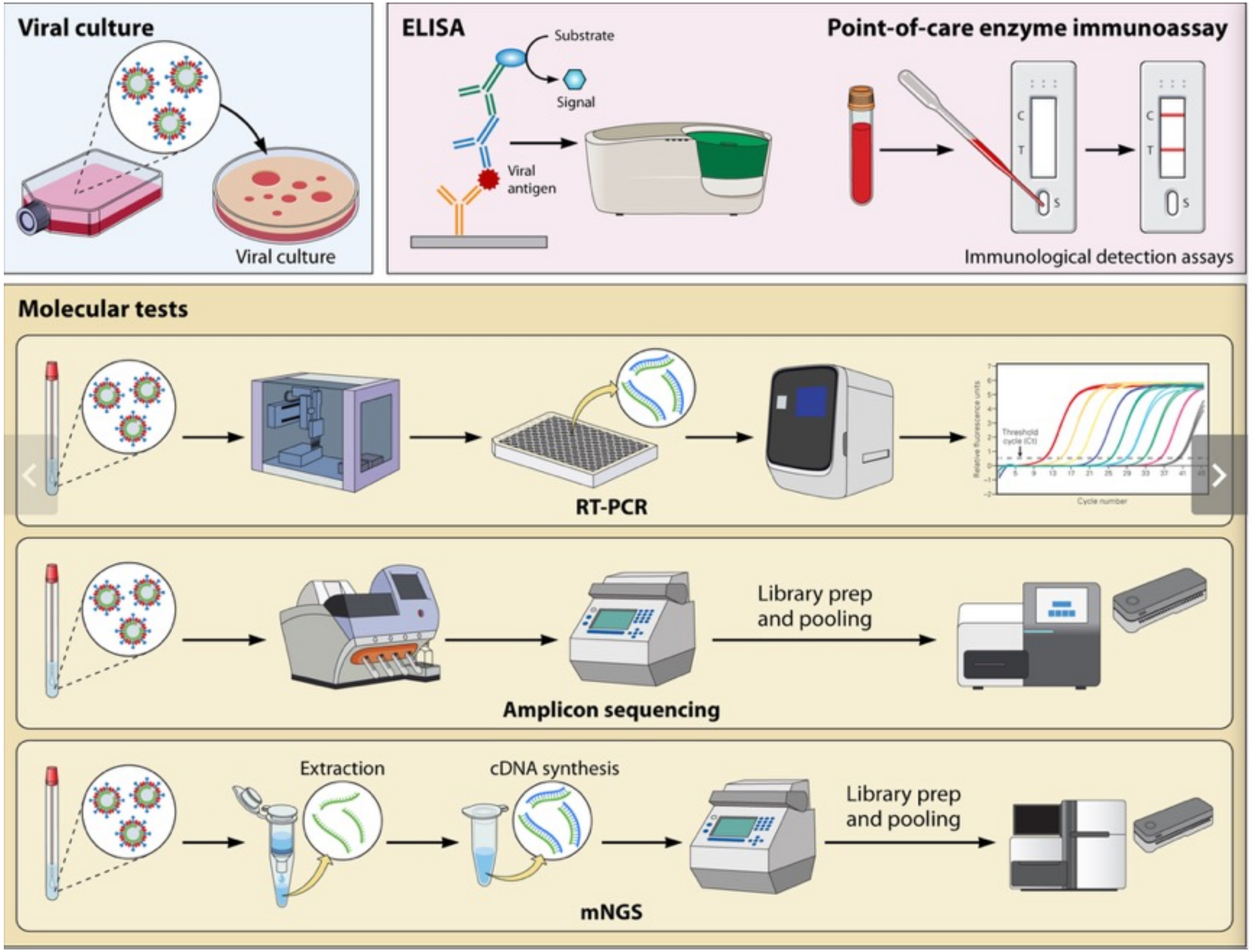
Comparing Benchtop Sequencing Platforms

1st Gen 2nd Gen 3rd Gen

Instrument	GeXp	SeqStudio	Pyromark	MiSec	NextSeq	MinION*	GridION
Platform	Sciex Sanger	Thermo Fisher Sanger	Illumina NGS	Illumina NGS	Illumina NGS	ONT Single	ONT Single
Sequencing Generation	First	First	Second	Second	Second	Third	Third
Output Gb/run	2-800 Kb	2-800 Kb	2 Gb	15 Gb	120 Gb	10 Gb	50 Gb
Read Length (bp)	400-900	400-900	2 X150	2 X 300	2 X150	>> 10K	>> 10K

*MinION is the only portable sequencer, all the others are benchtop sequencers

Pathogen (virus) Diagnosis



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Genomics

- DNA sequencing and the capacity to investigate the genome of host and pathogen populations have become an essential part of biomedical research, unlocking unique information about why some individuals or populations develop disease while others do not and how pathogens evolve to cause disease and counteract medical interventions
- There is exponential growth in the interest and implementation of genomics research in Africa. However, most of the sequencing of samples collected in Africa is done outside of the continent.

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State of play for Genomics in SSA

- > 90% of sequencing of samples from Africa is not done in Africa, (source: H3 Africa meeting 2023)
- Urgent need to make good use of the available Biobank resources and other Biorepositories in the unit and all of Africa

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Challenges/ Opportunities - I

- Managing genomics technology platforms in LMIC
- Secure resources and platforms that are science driven
- Organise training and attain international accreditation
- Setting good operating procedures, SOP's and workflows

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Challenges/ Opportunities - II

- Establish a state of the art Genomics core,
- MRC G initiative, to train and develop scientists, and be a Sub - Sahara Africa (SSA) Genomics Hub and COE
- Strive to improve and evolve with current and **future** technologies
- Engage stakeholders (user and management and external groups) at all stages, …

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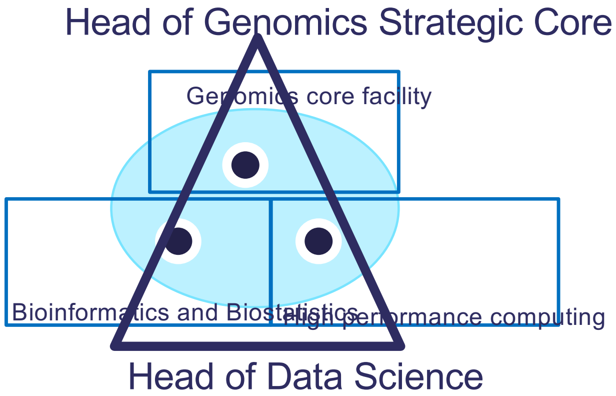
Aligning with the unit vision – simplified network

- Understanding the molecular pathway to diseases
- Building capacities and capabilities



Aligns with the school mission: Improve health and health equity in the UK and worldwide; working in partnership to achieve excellence in public and global health research, education and translation of knowledge into policy and practice.

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Establishing a Genomics core platform at MRCG @LSHTM

Genomics Facility

- 1st, 2nd and 3rd Generation sequencers
- Molecular equipment

- Platform Staffs
- Project Staffs
- MPhil and PhD Students
- Interns
- Fellows

Biobank

- Archived samples
- Current projects
- Quality control

MB and genomics

- Molecular core
- Infrastructure
- Access
- Governance

Data management and Computation

- Data management
- High performance computing (HPC)

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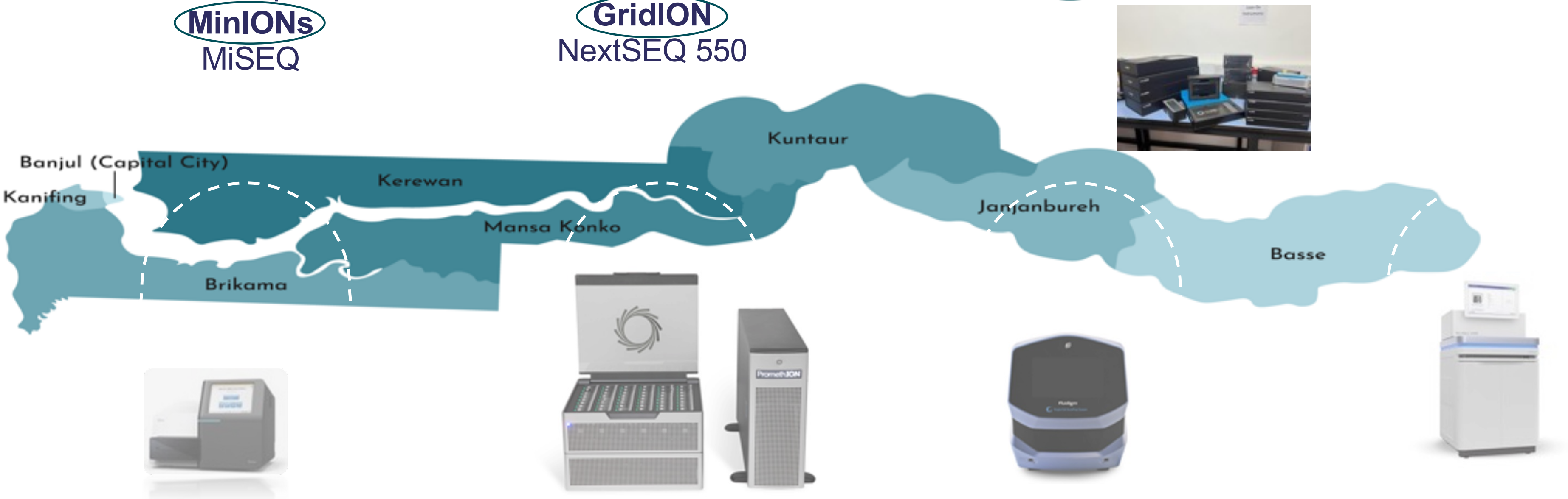
Investments in genomics platforms at MRCG @ LSHTM

2017
GeXp
MinIONs
MiSEQ

2019
SeqStudio
GridION
NextSEQ 550

2021
MK1Cs

2023
NovaSEQ 6000



2018
Pyromax48

2020
PromethION
NanoString Systems
iScan

2022
Fluidigm C1, JUNO,
& BioMark HD
10X

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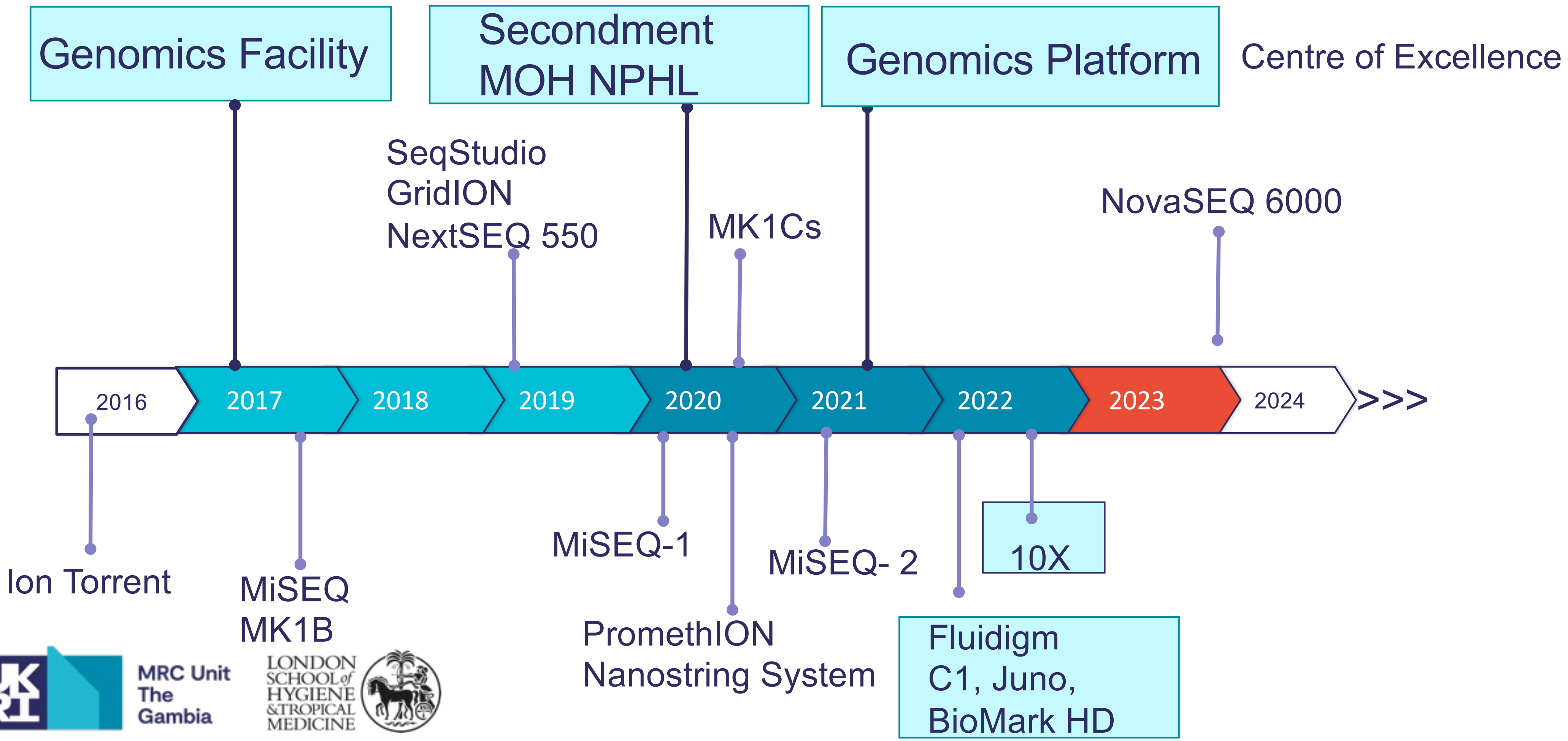
My Genomics journey at MRCG at LSHTM



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Collaborations and Partnerships

- Bubacarr Bah
- Badou Gaye
- Nureeden Muhamed

Genomics & Data Science

West Africa

Martin Antonio

- ASLM
- WAHO
- WHO AFRO
- WANETAM

- Anna Roca
- Alfred Ngwa
- Jahangir Hossain
- Grant Mackenzie

Disease Control & Elimination

G S Core Platform

Nutrition & Planetary Health

- Andrew Prentice
- Matt Silver
- Kris Murray

- Thushan de Silva
- Rachel Tribe
- Mahmod Maina
- Sam Sheppard
- Jose Ordovas-Montanes

External Partners

Vaccine & Immunity

- Beate Kampmann
- Jayne Sutherland
- Leopold Tientcheu

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Projects and grants

Thushan de Silva

Blood, Skin infections, Seasonal and Acute upper respiratory tract infections.

Bacterial and Viral Metagenomics, Single Cell Transcriptomics, AMR

Mariama Kujabi
Elina Senghore

Anna Rocca

- Effect of intra-partum azithromycin on the development of the infant microbiota.
- Sar-CoV-2 household transmission
- Sar-CoV-2 surveillance in WWR

Bacterial, Fungal and Viral Metagenomics, WGS, AMR

Umberto Dalessandro
Bakary Sanyang,
Pauline Getanda Kwamboka
Yayah Bah

Kris Murray

A social-ecological network approach to understanding zoonotic outbreak risk (SENZOR)
Zoonotic pathogens
The Gambia and Nigeria

Bacterial and Viral Metagenomics, WGS

Anise Happi, Nigeria
David Redding, UK
Almudena Mari Saez, France
Johanna Hanefeld, Germany

Matt Silver

PRECISE Spontaneous Pre-Term Birth (PRECISE-SPTB)
A Multi-OMICS study

Metagenomics, Transcriptomics, Epigenetics, GWAS, Proteomics

Sambou Suso
Rachel Tribe, UK
Geoffry Omouse, Kenya

Jahangir Hossain

Genomic Epidemiology and Transmission of Campylobacter in Africa (GETCampy-Africa)

Bacterial WGS, Metagenomics,

Martin Antonio
Isidore Bonkoungou, Burkina
Akosua Karikari, Ghana
Samuel Sheppard, UK

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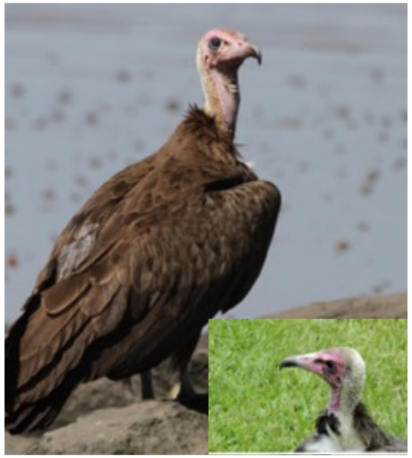
ORG One Project: Conservation Genomics

Critically Endangered vultures in The Gambia

Hooded Vulture



Rüppell's Vulture

White-backed Vulture

High-quality draft reference genomes

Improve conservation efforts with information on:

- species identity
- genetic disease risk
- how an organism has evolved.

One Zoo CDT Project

Pathogen pollution in critically endangered vultures in The Gambia

Stephanie Key — *PhD student*

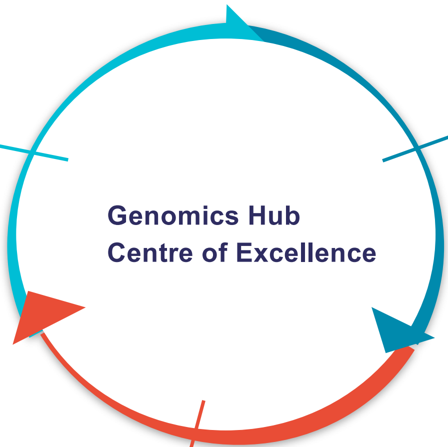
Kris Murray — *Co Supervisor*

The platform genomics capacity building model

Phase 1

Training and Support

- Hand-on laboratory training
- On-site and at MRCG
- Bioinformatics training at MRCG
- Access to High Performance Computing



Phase 2

Coaching and Mentoring

- Networking
- Regular on-line training
- Advice on training courses,
- Grants and fellowship

Phase 3

Monitoring, Evaluation, and Learning, Activity

- Site visits
- Support/ training on accreditation/certification
- External quality program
- Learning and accountability

THE LANCET
Global Health

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Simple and structured model to build sequencing capacity in west Africa

Abdoulie Kanteh • Jarra Manneh • Bakary Sanyang • Mariama A Kujabi • Haruna S Jallow • Dabiri Damilare K • Sainabou Laye Ndure • Abdul Karim Sesay • Show less

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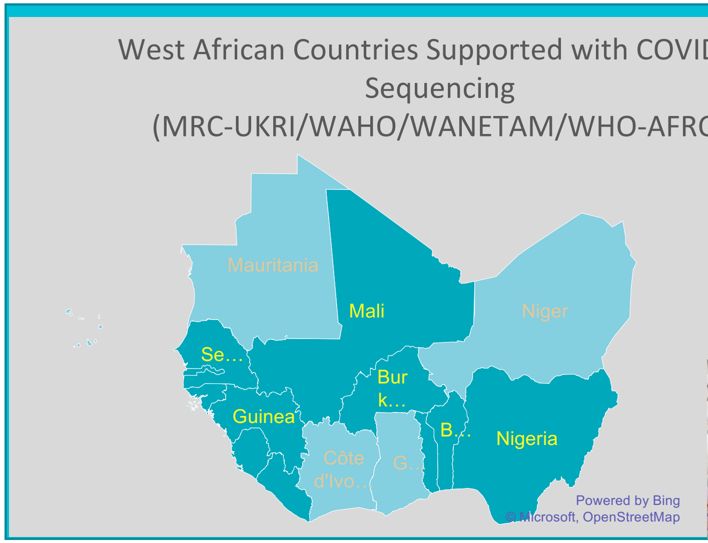
Building sequencing capacity to enhance surveillance of COVID-19 pandemic



Capacity Building Training in the last 18 months

Dakar, Senegal,	MRC UKRI (3 labs)
Cotonou Benin,	WHO
Bissau, Guinea Bissau,	MRC UKRI
Freetown, Sierra Leone,	MRC UKRI, WAHO
Nairobi, Kenya STPB,	MRC UKRI
Bamako, Mali,	WANETAM, MRC UKRI (2 labs)
Lagos, Nigeria,	WANETAM
Monrovia, Liberia	WAHO
Conakry, Guinea	WAHO
Bissau, Guinea Bissau	WAHO
Bobo, Burkina Faso,	MRC UKRI
Lome, Togo,	WAHO

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Platforms in the facility

Illumina platform (NGS)

MinION Mk1B MinION Mk1C MinION Mk1D

Nanopore platform

1st Gen. Sanger, FA

Pyromark 48

Life Technologies
Ion Torrent

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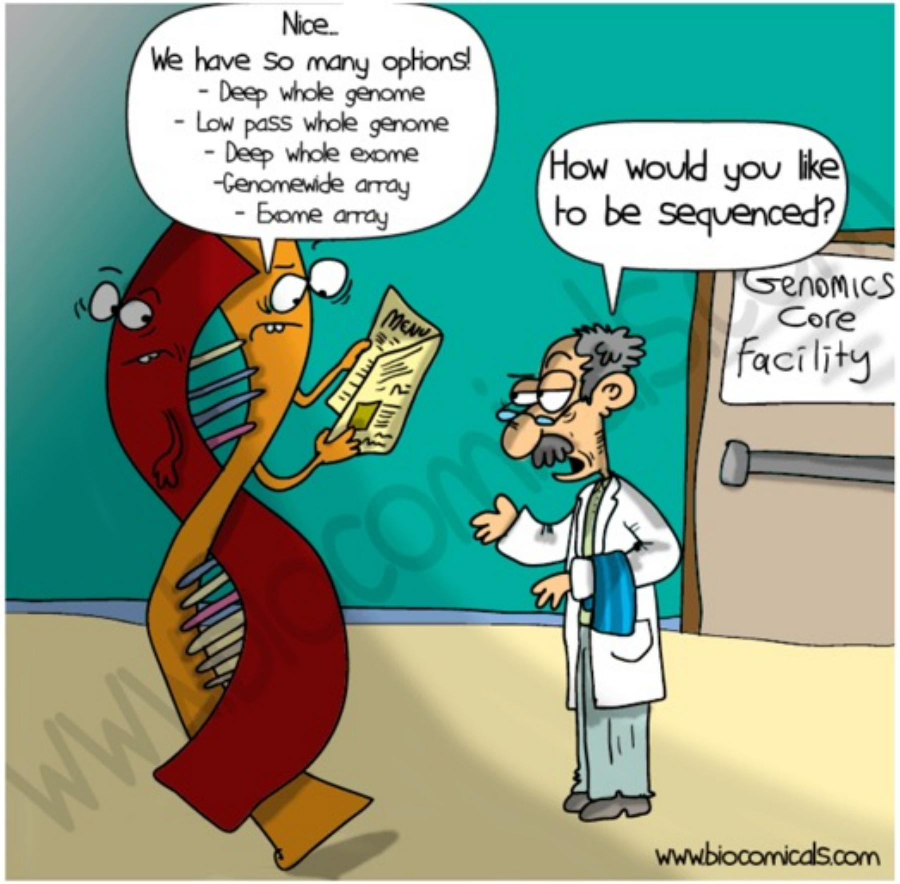
Would You Sequence your Genome



If Yes, why
If No, why not

Explain differences between:
RT PCR, Microaaray, 1st gen, 2nd gen and 3rd gen sequencing
In terms of throughput, plexing, cost, speed and access

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